

WBSEDCL 33/11 KV Substation

DIPLOMA IN ELECTRICAL ENGINEERING

A PROJECT REPORT

Submitted by

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CERTIFICATE

This is to certify that the project entitled ‘**WBSEDCL 33/11 KV SUB STATION**’ submitted to the Brainware University for the award of the **DIPLOMA IN ELECTRICAL ENGINEERING** is a Bonafide work carried out by “**M.VAMSHI KUMAR**” under supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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Place: Barasat, Kolkata.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **WBSEDCL** and the staff of the 33/11 KV Substation in Barasat for allowing us to visit their facility and providing us with an in-depth understanding of its operations. Their guidance and detailed explanations of the substation's components and functions were invaluable to this learning experience.

I am also grateful to Brainware University for organizing this educational visit as part of our curriculum, enabling us to gain practical insights into the power distribution network. I extend my heartfelt thanks to our professors and mentors for their constant support and encouragement, as well as for equipping us with the theoretical knowledge that enhanced our understanding during the visit.

Finally, I would like to acknowledge my peers for their collaboration and active participation, which made this experience both engaging and enriching.

ABSTRACT

This report details an educational visit to the WBSEDCL 33/11 KV Substation located in Barasat, conducted on December 2, 2024, as part of the academic curriculum at Brainware University. The substation serves as a critical node in the power distribution network, stepping down voltage from 33 KV to 11 KV for local distribution. During the visit, key components such as power transformers, circuit breakers, isolators, lightning arresters, and control panels were observed, along with their applications in ensuring efficient and reliable power delivery. The report highlights the practical understanding gained about substation operations, safety mechanisms, and the challenges in maintaining uninterrupted power supply. This hands-on experience bridged theoretical knowledge with real-world applications, fostering a deeper appreciation for the complexities of power systems.

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INTRODUCTION

- On December 2, 2024, I had the opportunity to visit the **WBSEDCL 33/11 KV Substation** located in **Barasat** as part of an educational visit organized by **Brainware University**. This visit was aimed at bridging the gap between theoretical knowledge and practical applications by observing firsthand how an electrical substation operates. Substations are vital in the transmission and distribution of electricity, ensuring that power reaches end-users efficiently and reliably.
- The **WBSEDCL 33/11 KV Substation** plays a critical role in the power distribution network, stepping down high voltage from 33 KV to 11 KV, making it suitable for local distribution. This facility is equipped with various advanced components and safety mechanisms to ensure uninterrupted and safe electricity delivery. Visiting such an establishment provided an invaluable opportunity to understand the intricate operations involved in managing and distributing electricity effectively.
- During the visit, I observed various components of the substation, including power transformers, circuit breakers, lightning arresters, and relay systems. Each component serves a specific purpose, contributing to the overall reliability and safety of the electricity distribution process. Furthermore, the visit offered insights into the challenges faced by utility providers and the solutions implemented to address them.
- This report documents the observations made during the visit, including an overview of the substation, the equipment observed, their applications, and the key learnings derived from this experience. The exposure gained from this visit has deepened my understanding of power systems and highlighted the importance of substations in maintaining a stable and efficient power grid.

OVERVIEW OF THE SUBSTATION

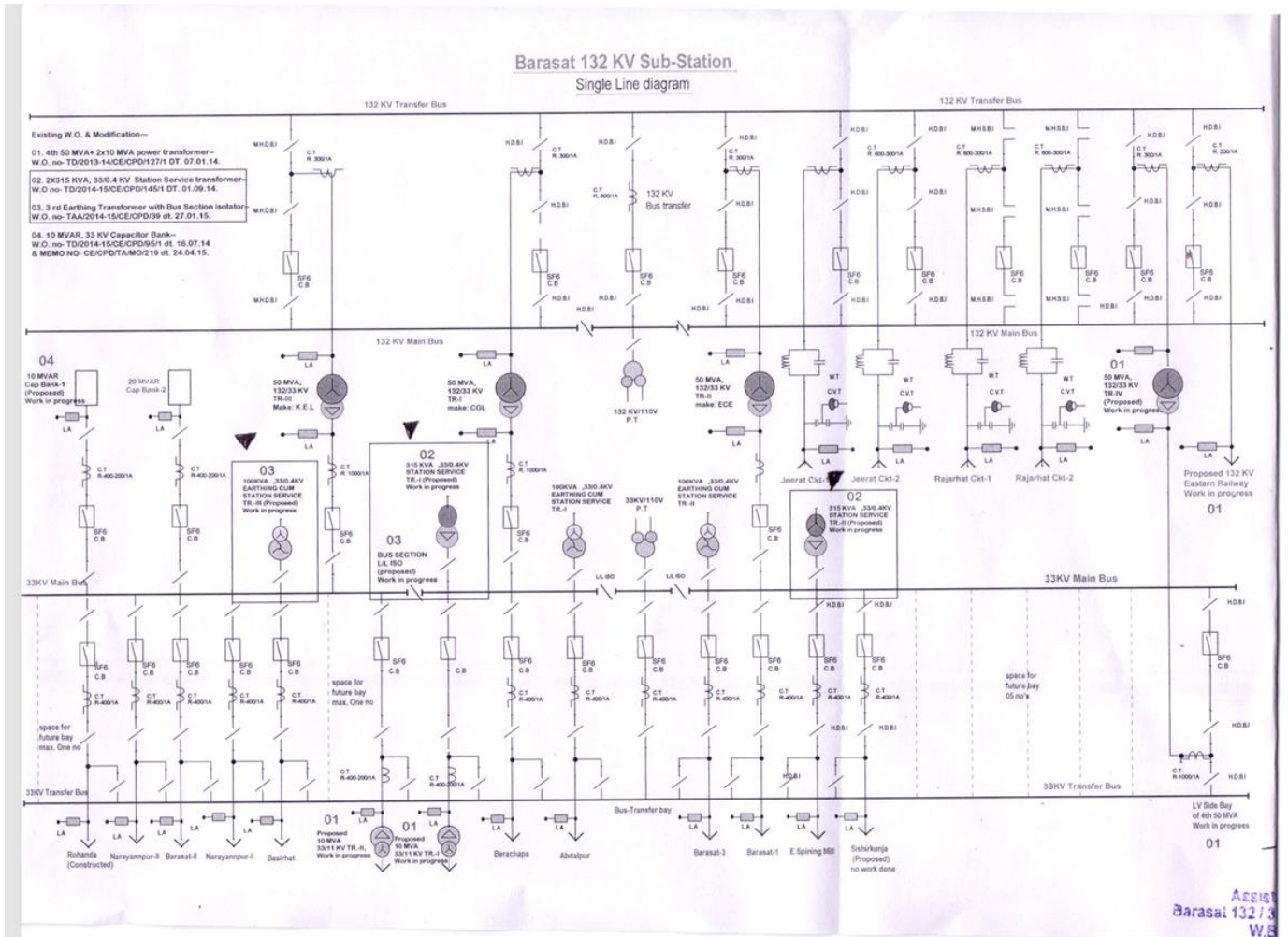
- The WBSEDCL 33/11 KV Substation in Barasat plays a pivotal role in distributing electricity to residential, commercial, and industrial consumers in the region. It functions as a critical intermediary point in the power distribution network, stepping down high voltage electricity received from the transmission grid (33 KV) to a lower voltage level (11 KV) that is suitable for local distribution. This ensures that electricity can be delivered safely and efficiently to end-users.
- The substation operates with the help of advanced electrical equipment and automation systems designed to handle high-voltage power safely and reliably. The layout of the substation includes outdoor installations for large equipment such as power transformers, circuit breakers, and isolators, as well as an indoor control room housing monitoring systems and relay panels. These components work in unison to manage the flow of electricity, monitor operational parameters, and protect the network from faults.
- The substation is also equipped with modern safety mechanisms, including lightning arresters and earthing systems, which are essential for protecting both equipment and personnel. In addition to power distribution, the facility plays a key role in maintaining voltage stability and improving the power factor through the use of capacitor banks.
- By observing the operations of this substation, it becomes evident how crucial it is in ensuring uninterrupted power supply to the region. The combination of robust infrastructure and skilled management makes this facility a cornerstone of the local power distribution system.

EQUIPMENT AND COMPONENTS OBSERVED

During the visit, we have observed the following key components and equipment:

- **Power Transformers:** Used for stepping down the voltage from 33 KV to 11 KV.
- **Circuit Breakers:** Installed to interrupt fault currents and protect equipment during overloads or short circuits.
- **Isolators:** Devices used for isolating a portion of the circuit for maintenance.
- **Current Transformers (CTs) and Potential Transformers (PTs):** Used for measurement and protection purposes.
- **Lightning Arresters:** Installed to protect equipment from voltage surges caused by lightning strikes.
- **Control Panels:** Located in the control room, these panels are used to monitor and control various operations in the substation.
- **Bus Bars:** Metal strips used to carry and distribute power within the substation.
- **Earthing Systems:** Ensures the safety of equipment and personnel by dissipating fault currents into the ground.
- **Capacitor Banks:** Used for power factor correction and voltage regulation.
- **Relay Systems:** Provide protection by triggering circuit breakers in case of faults.

SINGLE LINE DIAGRAM OF BARASAT 132 KV SUB-STATION



BARASAT 132 KV SUB-STATION EQUIPMENT DETAILS

Transformers

Equipment Type	Rating	Voltage Level	Identifier	Purpose / Function
Power Transformer	50 MVA	132/33 kV	Jeerat Ckt-1, Ckt-2, Rajarhat Ckt-1, Ckt-2, Proposed Railway, WBSEB	Main power supply to 33 kV level from 132 kV
Station Service Transformer	2 x 20 MVA	132/33 kV	ID: 02	Service power for substation operation
Station Service Transformer	3150 kVA	33/0.433 kV	ID: 02	Internal low-voltage power requirements
Earthing Transformer	1000 kVA	33/0.433 kV	ID: 03	Neutral grounding and fault protection
Power Transformer (Proposed)	50 MVA	132/33 kV	ID: 01	For future line expansion
Station Transformer (Proposed)	3150 kVA	33/0.433 kV	ID: 02	For future station service
Earthing Transformer	3150 kVA	33/0.433 kV	ID: 03	For grounding during SLG faults

Capacitor Banks

Equipment Type	Rating	Voltage Level	Identifier	Status
Capacitor Bank	20 MVAR	33 kV	ID: 03	Work in Progress

Capacitor Bank	10 MVAR	33 kV	ID: 04	Proposed
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Bus Configuration

Bus Type	Voltage Level	Components / Function
Main Bus	132 kV	Includes transformers, feeders, isolators, breakers
Transfer Bus	132 kV	Backup line connection
Main Bus	33 kV	Load distribution and transformer connections
Transfer Bus	33 kV	Standby operation and bypass routes
Bus Section	33 kV	ID: 03 - For sectionalizing the 33 kV bus

Instrument Transformers

Type	Rating	Voltage Level	Location
Potential Transformer (PT)	132 kV/110 V	132 kV	Main bus (metering/protection)
Current Transformers (CTs)	-	132 kV & 33 kV	Near breakers & isolators

Switchgear

Equipment	Symbol	Voltage Level	Function
Circuit Breakers	C.B.	132 kV / 33 kV	Interrupt fault or load current
Isolators	I.S.	132 kV / 33 kV	Manual disconnection for maintenance
Lightning Arresters	L.A.	132 kV / 33 kV	Over-voltage protection

Feeders and Lines

Feeder Name	Voltage	Status
Jeerat Ckt-1	132 kV	Active
Jeerat Ckt-2	132 kV	Active
Rajarhat Ckt-1	132 kV	Active
Rajarhat Ckt-2	132 kV	Active
Barasat-1 to Barasat-3	33 kV	Active
E-Spinng Mills	33 kV	Active
Berachapa, Adalpukur	33 kV	Active
Sakuripukur	33 kV	Proposed

Auxiliary Sections & Notes

Description	Voltage / Type	Details
Bus Transfer Bay	33 kV	Used to switch loads between buses
Control Room Connections	33 kV	For feeder control (implied)
Space for Future Bay	132 kV / 33 kV	Reserved for expansion
132/110 V PT	132 kV	For metering and protection

BARASAT 132 KV SUBSTATION – SINGLE LINE DIAGRAM COMPONENTS

1. Power Transformers (e.g., 50 MVA, 132/33 kV)

- **Function:** Step down voltage from 132 kV to 33 kV for distribution.
- **Use:** Allows transmission over long distances (high voltage) and distribution at safer, usable voltages (low voltage).

2. Circuit Breakers (C.B.)

- **Function:** Protect equipment by interrupting fault currents.
- **Use:** Ensures safe operation by isolating faulted sections of the system during overcurrent or short circuit.

3. Current Transformers (C.T.)

- **Function:** Measure current and provide inputs to relays and meters.
- **Use:** Step down high current to measurable levels for monitoring and protection.

4. Potential Transformers (P.T. or V.T.)

- **Function:** Step down voltage to a safe level for measurement and protection.
- **Use:** Feeds voltage signals to relays and meters for monitoring and control.

5. Lightning Arresters (L.A.)

- **Function:** Divert high-voltage surges to the ground (e.g., from lightning).
- **Use:** Protects transformers and other sensitive equipment from overvoltage.

6. Isolators (HDBI, MDBI)

- **Function:** Manually disconnect sections for maintenance.
- **Use:** Operated under no-load conditions to ensure safety during service work.

7. Busbars (132 kV Bus, 33 kV Main Bus, Transfer Bus)

- **Function:** Conduct and distribute power between incoming and outgoing circuits.
- **Use:** Provide flexibility in operation and isolation during faults or maintenance.

8. Earth Switches

- **Function:** Ground sections of busbars or lines.
- **Use:** Provide safety during maintenance by ensuring no voltage is present.

9. Reactors (e.g., 10 MVAR, 20 MVAR)

- **Function:** Control voltage rise during light load conditions.
- **Use:** Improve voltage stability and reduce transmission losses.

10. Capacitor Banks (e.g., 10 MVAR, 20 MVAR)

- **Function:** Improve power factor and voltage regulation.
- **Use:** Compensate for lagging power factor due to inductive loads.

11. Bus Coupler / Transfer Bay

- **Function:** Connects sections of busbars for load sharing or maintenance.
- **Use:** Enhances reliability and flexibility in power flow control.

12. Station Auxiliary Transformer

- **Function:** Supply power to substation equipment like lights, relays, etc.
- **Use:** Derives power from the main transformer for station use.

13. Relays and Protection Panels (not visibly detailed but implied)

- **Function:** Detect abnormal conditions and command breakers to operate.
- **Use:** Essential for automating fault detection and protection.

APPLICATIONS OF SUBSTATION COMPONENTS

Each component observed during the visit plays a crucial role in ensuring the substation's smooth functioning:

- **Power Transformers:** Enable the efficient transfer of electricity over long distances by stepping down voltage to safer levels for local distribution. These transformers also help in reducing energy losses during the transmission of electricity.
- **Circuit Breakers and Relays:** Protect the system from damage due to overloads, faults, or short circuits. They act as the first line of defense by quickly isolating the faulty section to prevent further damage to the network.
- **Current and Potential Transformers:** Provide accurate measurements for operational monitoring and fault analysis. These transformers ensure the reliability of protective relays and metering devices by scaling down high current and voltage levels to manageable values.
- **Lightning Arresters:** Protect expensive equipment from lightning-induced voltage surges. These devices divert the excess voltage safely into the ground, safeguarding sensitive electrical components.
- **Capacitor Banks:** Improve system efficiency by minimizing reactive power losses. They help in voltage stabilization and reducing the strain on power generation units, making the system more reliable.
- **Bus Bars:** Serve as distribution points for efficiently routing power to various feeders. Their robust design allows for the simultaneous transmission of electricity to multiple circuits, enhancing the operational flexibility of the substation.
- **Isolators:** Enable maintenance and repair work by safely isolating sections of the circuit. This ensures that repairs can be conducted without disrupting the entire power system.
- **Control Panels:** Serve as the operational hub of the substation, allowing real-time monitoring and management of electrical parameters. They enable quick decision-making in response to system anomalies, ensuring seamless power delivery.

KEY LEARNINGS FROM THE VISIT

The visit provided valuable practical knowledge about the functioning and importance of a substation in the power distribution network. Key learnings include:

- Understanding how electricity is stepped down from high voltage to medium voltage for distribution.
- The importance of protective devices like circuit breakers, relays, and lightning arresters in maintaining system reliability.
- The significance of earthing systems for ensuring personnel safety and equipment protection.
- The role of automation and control systems in monitoring and managing substation operations.
- Insights into the challenges faced by utility providers in ensuring uninterrupted power supply.

Photos that our group has captured in the visit, are posted below: -





CONCLUSION

The visit to the WBSEDCL 33/11 KV Substation in Barasat was an enriching experience, bridging theoretical concepts with real-world applications. It highlighted the critical role substations play in the power distribution network and underscored the importance of each component in maintaining system stability and reliability. Such practical exposures are invaluable for engineering students, providing a deeper understanding of the power sector and inspiring future contributions to the field.